

Avery Carter

Candidate ID: CAND_006

Education: B.Tech in Computer Science and Engineering

Experience: 2.8 years

Professional Summary

Interactive software engineer focused on real-time entertainment applications and 3D interactive experiences. Experienced in building responsive gameplay systems, believable virtual characters, physics-driven interactions, networked experiences, and performance-sensitive applications. Strong background in object-oriented software design and collaborative source-control workflows.

Technical Strengths

Area	Level
Systems Programming	Advanced
Object-Oriented Software Design	Advanced
Interactive 3D Development	Advanced
Real-Time Simulation	Advanced
Virtual Character Behaviour	Advanced
Interactive Physics & Collision Systems	Advanced
Game Engine Development	Advanced
Networked Multiplayer Systems	Intermediate
Scripting & Automation	Advanced
Version Control / Collaborative Development	Advanced
Algorithm Design	Advanced
Performance Optimization	Advanced

Professional Experience

Interactive Software Engineer — NovaPlay Interactive | 2.8 years

- Built real-time interactive systems for PC and console entertainment products.
- Developed gameplay mechanics, movement systems, object interactions, and responsive control systems.
- Created virtual-character decision logic for opponents and non-player entities.
- Implemented realistic movement, collision responses, and physical interactions between objects.
- Worked on online co-operative features involving client/server communication and state synchronization.
- Profiled CPU and memory usage and optimized performance for stable real-time frame rates.
- Designed reusable components using object-oriented principles.
- Maintained shared source code using Git-based development workflows.

Selected Projects

Neon District — 3D Action Adventure

Created a third-person interactive environment with character movement, enemy decision-making, inventory mechanics, collision handling, and dynamic environmental interactions.

Velocity Arena — Online Racing Prototype

Built a networked racing prototype with vehicle handling, real-time synchronization, collision behaviour, and competitive multiplayer features.

RoboLab Simulation

Developed a simulation involving autonomous virtual agents navigating an environment and responding to obstacles and changing conditions.

Additional Experience

Regularly worked with debugging tools, profiling utilities, collaborative code reviews, branching workflows, and iterative development practices. Comfortable learning unfamiliar libraries and adapting existing systems to new gameplay requirements.

Certifications & Activities

- Real-Time Interactive Development Certificate
- Participated in two university game-development competitions
- Open-source contributor to small interactive-software projects